2. Introduction and Background

Student grade management is an important academic task involving the recording, calculation, and retrieval of student performance. This project develops a console-based application that manages student records in memory using Python. Week 1 focuses on implementing the core features, including grade calculation, a menu-driven interface, and operations to add, view, search, update, and delete student records.

3. Objectives of the Project (Week 1)

- Develop a menu-driven application for managing student records.
- Implement Add, View, Search, Update, and Delete operations.
- Calculate grades from percentage scores.
- Build a reusable bordered terminal interface.
- Validate roll numbers and marks.
- Test all CRUD operations.

4. Problem Statement / Driving Question

How can student records, including marks, percentages, and grades, be efficiently managed using a simple terminal-based application?

5. Methodology (Week 1 Activities)

1. Data Model & Storage:

- Used an in-memory STUDENTS dictionary to store student records and a fixed SUBJECTS list for consistent data handling.

2. Grade Calculation Logic:

- Implemented calculate_grade() to assign letter grades and get_report() to calculate total marks, percentage, and grades

3. Terminal UI & Menu System:

- Developed reusable box-drawing functions for the interface, implemented a menu with Add, View, Search, Update, Delete, and Exit options, and added input validation for roll numbers and marks.

6. Expected Outcomes / Deliverables (Week 1)

- A working console application supporting Add, View, Search, Update, and Delete of student grade records
- A functioning calculate_grade() module producing consistent letter grades from percentage scores
- A bordered, readable terminal UI built from reusable box-drawing utilities
- Input validation for duplicate roll numbers, invalid marks, and non-numeric entries
- Week 1 project report